action is a function that maps a state, *s*, to a new state, *s′*. An action may be stochastic, such that it maps a state, *s*, probabilistically to a new state, *s′*. There may be some probability distribution over the set of possible new states from which one is selected. The action-space is the set of all possible actions for a particular state.

- The environment is the source of states, actions, and rewards. If we're building an RL algorithm to play a game, then the game is the environment. A model of an environment is an approximation of the state space, action space, and transition probabilities.
- Rewards are signals produced by the environment that indicate the relative success of taking an action in a given state. An expected reward is a statistical concept that informally refers to the long-term average value of some random variable *X* (in our case, the reward), denoted $E[X]$. For example, in the *n*-armed bandit case, $E[R|a]$ (the expected reward given action *a*) is the long-term average reward of taking each of the *n*-actions. If we knew the probability distribution over the actions, *a*, then we could calculate the precise value of the expected reward for a game of *N* plays as $E[R|a_i] = \sum_{i=1}^{N} a_i p_i \cdot r$, where *N* is the number of plays of the game, p_i refers to the probability of action a_i, and *r* refers to the maximum possible reward.
- An agent is an RL algorithm that learns to behave optimally in a given environment. Agents are often implemented as a deep neural network. The goal of the agent is to maximize expected rewards, or equivalently, to navigate to the highest value state.
- A policy is a particular strategy. Formally, it's a function that either accepts a state and produces an action to take or produces a probability distribution over the action space, given the state. A common policy is the epsilon-greedy strategy, where with probability ε we take a random action in the action space, and with probability ε – 1 we choose the best action we know of so far.
- In general, a value function is any function that returns expected rewards given some relevant data. Without additional context, it typically refers to a state-value function, which is a function that accepts a state and returns the expected reward of starting in that state and acting according to some policy. The Q value is the expected